

Artificial Intelligence Based Clinical Decision Support Systems

Lecture Notes and Presentation Source Text

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Technology and Artificial Intelligence Literacy Training in Health Sciences, Akdeniz University, 16 September 2026

Abstract

This text is the source material for a session delivered within the Technology and Artificial Intelligence Literacy Training in Health Sciences programme. It begins with the definition of clinical decision support systems, sets out the stages through which such a system is built as software, examines why these systems fail in practice through independent external validation findings, and summarises the criteria that determine trustworthiness together with the regulatory framework. The development stages are mapped one by one onto the notebooks of the workshop held on 18 September 2026.

Key concepts: Clinical decision support system, software life cycle, external validation, calibration, positive predictive value, data leakage, explainable artificial intelligence, medical device software.

Session outline

Section	Key message
Introduction	Literacy in health begins with responsibility rather than technique
What a decision support system is	The system does not diagnose, it allocates clinical attention
How such a system is built	Twelve of thirteen stages correspond to a notebook in the workshop
Why these systems fail	A performance claim without external validation remains a claim
What makes a system trustworthy	Calibration, explainability and reporting standards are

Section	Key message
	assessed together
Regulatory framework	The absence of a dedicated statute is not an absence of liability
Bridge to the workshop	Generative AI writes code, the burden of assessment stays with the user

1. Introduction: the purpose of this session

Artificial intelligence literacy in health has a dimension that precedes technical skill. The party that bears the cost of an error is not the party that builds the system. The session is therefore organised around the relationship between how a decision support system is built and how it should be assessed. The two questions cannot be separated: a user who does not know the stages cannot see at which stage an error was introduced.

The coding work is largely delegated to generative artificial intelligence tools in the workshop of 18 September. What cannot be delegated is deciding which code to ask for, and inspecting the code that arrives.

2. What a clinical decision support system is

2.1. Definition and the five rights

A clinical decision support system is defined as a system that delivers the right information to the right person in the right format through the right channel at the right time, with the aim of improving a clinical decision. These five components are known in the literature as the five rights of clinical decision support [1, 2].

When the five components do not hold simultaneously, the system remains clinically ineffective even if it functions correctly in technical terms. Correct information delivered to the wrong user becomes noise, and correct information delivered after the decision point becomes only a record. Assessment therefore depends as much on how a system is embedded in the clinical workflow as on model performance.

2.2. Two families: knowledge based and data driven systems

Knowledge based systems rest on explicit rules derived from clinical guidelines. Drug interaction alerts, dose limit checks and allergy controls belong to this family. Their principal strength is that the justification for an alert is always readable, which makes them auditable and straightforward to version. Their weakness is that maintenance becomes difficult as the number of rules grows, and clinical nuance is not captured.

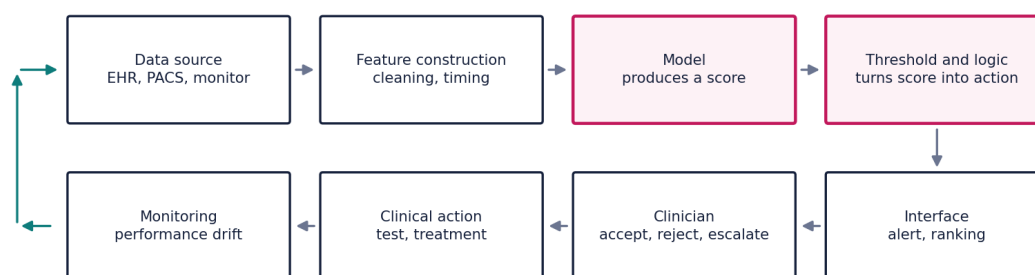
Data driven systems rely on models that learn patterns from historical patient data. Sepsis risk scores, readmission prediction and lesion marking in medical images belong to this group. Such systems can capture patterns that no one has been able to formulate explicitly. In return, what

they learn may be the recording habits or device infrastructure of the institution where they were trained, and performance degrades when the system is transferred elsewhere.

Most systems in the field are hybrid: a learned score is placed behind a rule based threshold. The way responsibility is established differs between the two families, and the distinction reappears in regulatory assessment.

2.3. System anatomy

A clinical decision support system can be viewed as a chain of eight links. The chain begins with the data source and continues with feature construction, the model, the threshold and decision logic, the interface, the clinician decision, the clinical action, and finally monitoring and feedback.



Published performance belongs to the third link. Clinical benefit arises after the fourth.

Figure 1. The eight links of a clinical decision support system and the feedback loop.

Performance reported in publications usually belongs to the third link alone, that is, to the model itself. Clinical benefit is determined from the fourth link onward. Model performance and clinical benefit are not interchangeable.

2.4. The position of the clinician

The output of a decision support system is not a diagnosis but an allocation of attention. The system does not declare that a particular condition is present; it proposes that the patient be prioritised. The difference matters both clinically and legally.

In the United States, Section 520(o)(1)(E) of the Federal Food, Drug, and Cosmetic Act, added by the 21st Century Cures Act, sets out four criteria under which decision support software falls outside the device definition. One of them is that the health care professional is able to independently review the basis for the recommendation [3, 4]. Explainability in that framework is not merely an ethical preference but the threshold determining whether the software is regulated as a device.

3. How such a system is built as software

Building a clinical decision support system can be divided into thirteen stages. The first twelve correspond to a notebook in the workshop of 18 September; the thirteenth lies outside the workshop and is discussed here with its rationale.

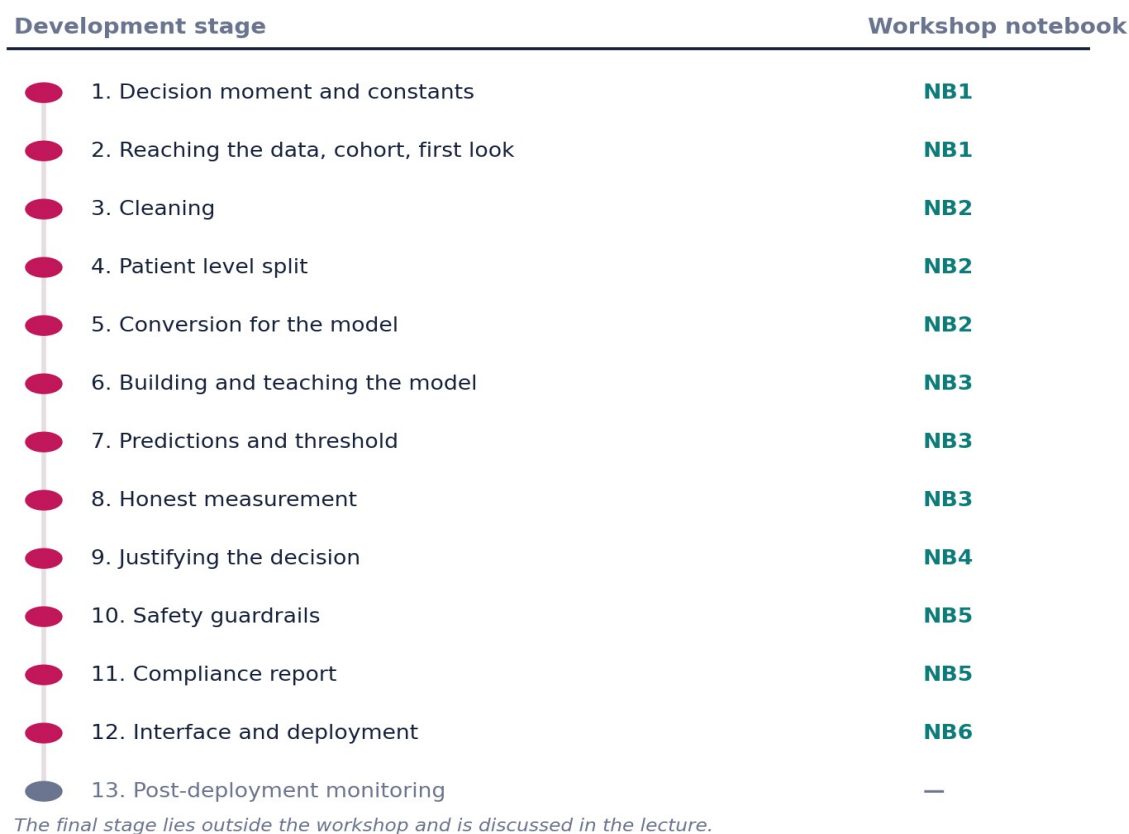


Figure 2. Development stages and the workshop notebooks they correspond to.

3.1. Decision moment, data and cleaning: stages 1 to 3

The first stage defines the decision moment and writes the constants of the program. The decision moment is the exact time at which the system produces output, and it draws the boundary of the information the system may use. In the shared scenario of the workshop that moment is six hours after admission to intensive care. The target threshold, the data source and the reproducibility number are defined here as well. Keeping these values at the head of the code and in one place is not a matter of style; in a clinical system thresholds are subject to audit and have to be visible.

The second stage reaches the data and defines the cohort. A cohort definition covers which patients the system leaves outside its scope as much as which it includes. In the shared scenario, stays shorter than six hours are excluded because those patients never reach the decision moment. The first defence against leakage is built here too: columns known only after the patient leaves, such as length of stay and time of discharge, are removed while the data is still being loaded.

The third stage is cleaning. Entirely empty columns, duplicated records and physiologically impossible values are dealt with. Where a value is impossible, emptying that cell is preferred to deleting the row; the rest of the patient record remains usable. Which value counts as impossible is a clinical decision.

3.2. Split and conversion: stages 4 and 5

The fourth stage splits the data into training and test groups, and the split must be made at patient level. Where a patient has more than one stay, a row level split distributes the same patient across both groups; the model recognises that patient and the test result shows a rise that does not exist in reality. The same problem arises in medical imaging and in physiological signals: two images or two signal segments from the same patient must not fall into separate groups either.

The fifth stage converts the data into the form the model can use. Non-numeric information is turned into numbers, missing values are filled and numeric columns are brought onto comparable scales. How these conversions are performed must be learned from the training group alone and applied unchanged to the test group. If the median used to fill missing values is computed across all the data, information about the test patients leaks into the learning process.

That is why the fifth stage follows the fourth. When the order is reversed the resulting error is silent: it raises no warning, lifts the test result somewhat and looks reasonable when the code is read.

3.3. Model and measurement: stages 6 to 8

The sixth stage builds and teaches the model. A simple and interpretable model is used first, and the reason is not performance but three properties. What a simple model looks at can be read, it produces probabilities, and it is less inclined to memorise on little data. A more complex model may beat it, and the yardstick for comparison remains in hand.

The seventh stage produces predictions and sets the threshold. The model must return probabilities rather than class labels. A model that returns labels directly has fixed the threshold at its own default, and that default is not a clinical decision. The clinical team sets the threshold: it is lowered where a miss is the costly error and raised where a false alarm is.

The eighth stage is measurement, and it receives more room in this lecture than any other. Six headings are reported together: discrimination with a confidence interval, calibration, sensitivity and specificity with the predictive values at the chosen threshold, the clinical translation of the operating point, the subgroup breakdown, and a comparison against a rule that learns nothing. The absence of any one of them renders the report uninterpretable.

3.4. Reasoning, safety and compliance: stages 9 to 11

The ninth stage justifies the decision. Overall reasoning shows which information the model depends on, and per patient reasoning shows what drove a single prediction. The clinician needs the second. The principal use of reasoning is error detection; it is not an assurance of correctness and carries no causal claim.

The tenth stage installs the safety guardrails. Three work together: the model abstains where the probability sits very close to the decision threshold; it produces no prediction for a patient unlike the training population; and it proposes no decision for a patient most of whose information is missing. The last is the most easily skipped, because the filling step closes every gap with average values and the model returns a confident number for a patient about whom nothing is known.

The eleventh stage is the compliance report. It records the purpose of the system, its user, its data, the performance measured, its guardrails, its known limitations, its regulatory position and the questions that remain open. The performance figures are drawn from the output of the measurement stage rather than typed by hand.

3.5. Interface and monitoring: stages 12 and 13

The twelfth stage covers the interface and deployment. An interface should not display a single number; the decision, the state of the guardrails and the reasoning are presented together. This is not only a design preference. Whether a health care professional can independently review the basis of a recommendation bears directly on the regulatory position of the software. Demographic attributes are not mixed into the reasoning list; such a contribution is a fairness audit finding rather than a reason to show a clinician.

The thirteenth stage is post-deployment monitoring and is not carried out in the workshop. Performance drift is noticed only where it is measured. Protocols, coding habits and patient profiles change over time while the model stays fixed, and performance erodes silently. Deployment without monitoring approaches deployment without validation.

3.5. Life cycle standards

The stages above align with international standards established for medical device software. IEC 62304 defines the life cycle processes for medical device software and assigns software to one of three safety classes, A, B or C, according to the harm a failure could cause; the required level of documentation, verification and testing follows from that class [10]. The only normative reference of the standard is ISO 14971, which governs risk management for medical devices, and the software specific risk management requirements are collected in Clause 7 of IEC 62304 [10, 11].

How the ISO 14971 process applies to the particular risks of machine learning is addressed in separate guidance. BS/AAMI 34971 describes how that process is applied to medical technology incorporating machine learning without modifying it, and addresses the risks that are elevated by, or unique to, such technology [12].

The prototype built in the workshop does not produce the documentation these standards require. The compliance report prepared at stage eleven does, however, show which questions that documentation would have to answer.

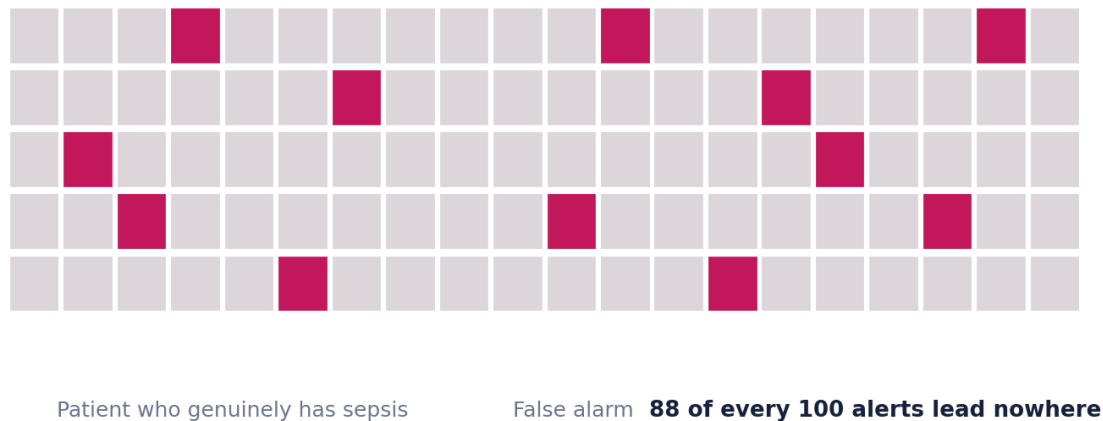
4. Why these systems fail

4.1. The Epic Sepsis Model case

The Epic Sepsis Model is a proprietary sepsis early warning model embedded in an electronic health record system used in hundreds of hospitals in the United States, and it generates automated alerts for clinicians. A team at Michigan Medicine externally validated the model on 38,455 hospitalisations belonging to 27,697 patients between 6 December 2018 and 20 October 2019 [5].

The findings were as follows. The area under the curve was 0.63 with a 95 percent confidence interval of 0.62 to 0.64, sensitivity was 33 percent, specificity 83 percent, positive predictive

value 12 percent and negative predictive value 95 percent. The range cited in the vendor documentation was 0.76 to 0.83 [5, 6].



Depiction of the 12 percent positive predictive value reported by Wong et al. (2021).

Figure 3. What a positive predictive value of 12 percent means on a shift roster.

A positive predictive value of 12 percent means that of every hundred alerts the system produces, eighty eight result in a patient being reviewed without cause. A specificity of 83 percent appears acceptable in a model report; the figure above is its equivalent in practice.

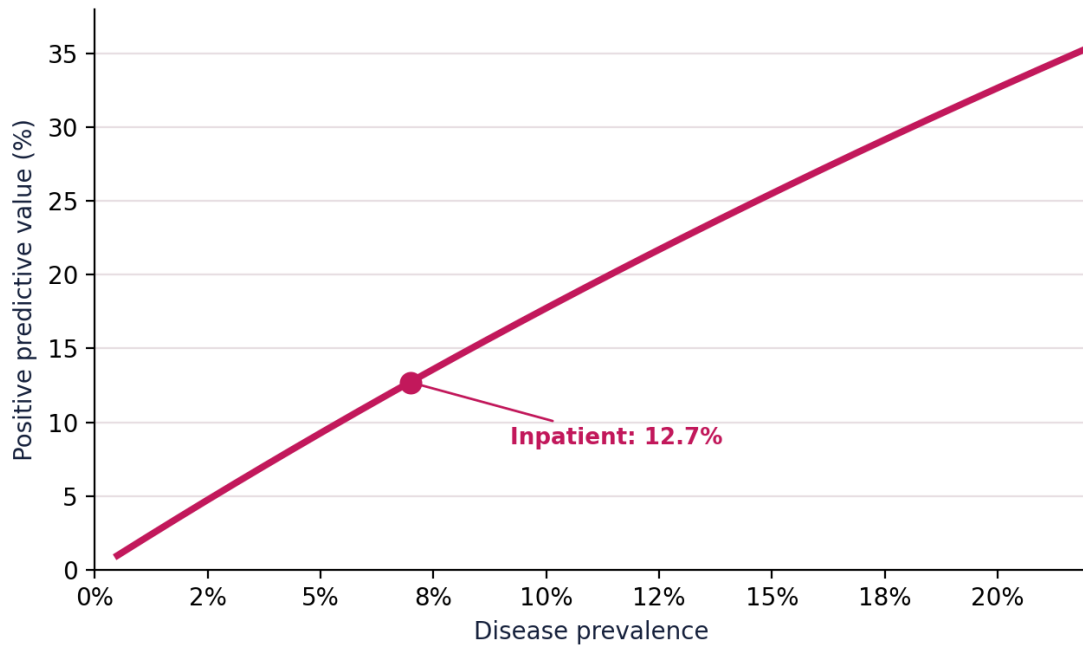
4.2. Three years later, in a different setting

The same product was reassessed across 145,885 encounters in two county emergency departments during 2023. The Sepsis-3 definition was applied and the alert was triggered at the threshold recommended by the vendor. Sensitivity was 14.7 percent, specificity 95.3 percent, positive predictive value 7.6 percent and negative predictive value 97.7 percent [7].

The lesson does not concern a particular product. No model carries its performance with it to the setting into which it is transferred. A different patient population, a different coding practice and a different workflow can each invalidate published performance.

4.3. The prevalence effect

Even when sensitivity and specificity are held constant, positive predictive value changes with the frequency of the condition in the clinic concerned. Under Bayes theorem, positive predictive value equals the product of sensitivity and prevalence divided by the sum of that product and the product of the false positive rate with the probability that the condition is absent.



Sensitivity 0.33 and specificity 0.83 held constant. The curve is computed from Bayes theorem.

Figure 4. Positive predictive value against prevalence, with sensitivity and specificity held constant.

In a system that screens for a rare condition, even near perfect specificity does not prevent a flood of false alarms. The practical rule that follows is that the first question a clinician should ask about a proposed system is not its area under the curve but the prevalence within the local patient population.

4.4. Alert fatigue and automation bias

The clinical cost of false alarms is cumulative. When most alerts prove unfounded, clinicians learn to dismiss alerts without reading them. This behaviour extends to correct alerts, so the system appears active while its protective effect approaches zero. The literature refers to this as alert fatigue.

The error that operates in the opposite direction is automation bias. A clinician may interpret the silence of the system as reassurance despite contrary examination findings, and accountability for a missed case becomes unclear. The two errors are not antidotes to one another and may appear simultaneously in the same system.

4.5. What the model learns may not be the disease

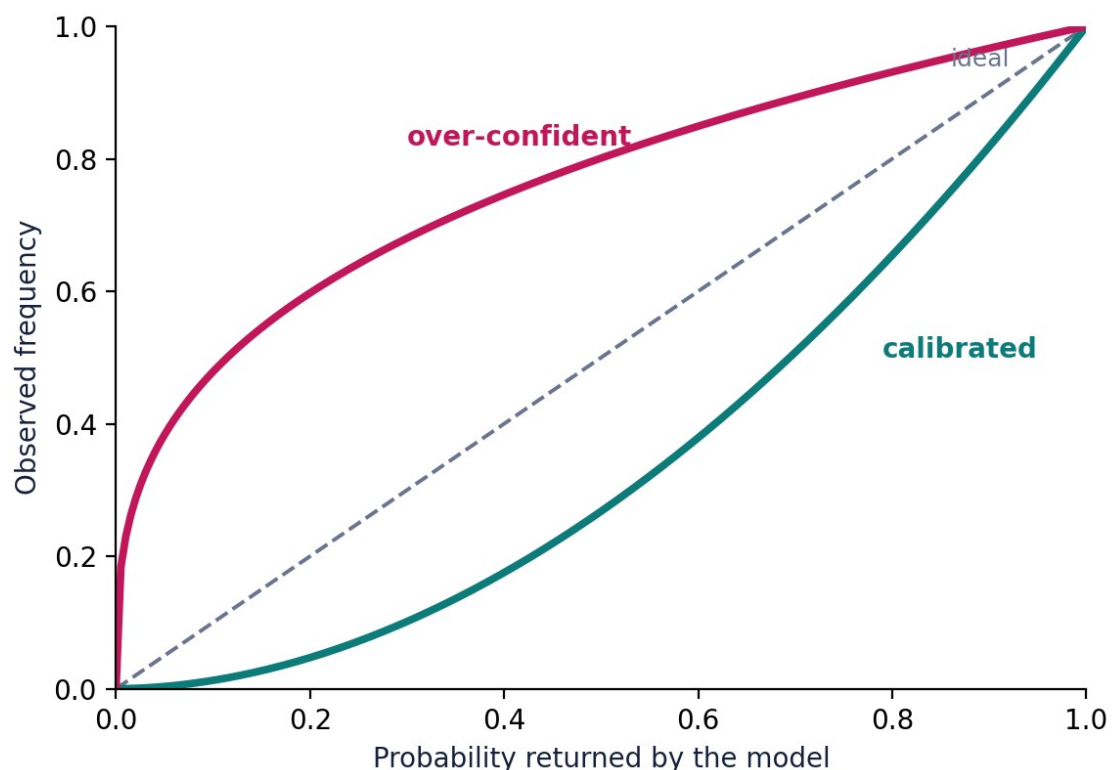
The technical reasons for lower than expected performance fall into four categories. The first is shortcut learning, where a model learns the marker a particular device prints onto an image rather than the pathology. The second is target leakage, where a consequence of the outcome enters the model as a feature. The third is distribution shift, where protocols or patient profiles change while the model stays fixed. The fourth is subgroup blindness, where overall performance appears acceptable while performance within a particular group is markedly lower.

None of these can be detected by inspecting an overall accuracy figure. Each becomes visible only when the data and the model are explicitly interrogated, and that interrogation is the central exercise of the workshop.

5. What makes a system trustworthy

5.1. Discrimination and calibration are not the same

Discrimination measures the ability of a model to separate an affected individual from an unaffected one and is a question of ranking. Calibration asks whether the probability the model reports corresponds to reality: among one hundred patients assigned a risk of 20 percent, does the event occur in twenty of them?



Schematic. The two models may share the same area under the curve.

Figure 5. Two models with the same discrimination and different calibration behaviour.

Two models may share the same area under the curve while one allows the clinician to set a meaningful threshold and the other assigns nearly every patient to the high risk group. Where a threshold is set for a clinical decision, calibration takes precedence over discrimination.

5.2. What explainability provides and what it does not

The strongest contribution of explainability to clinical assessment is error detection. If an inspection of the contributions shows that the decision is driven by the time at which the patient was registered, this is evidence of leakage. Marked differences across subgroups provide the first indication of a fairness problem.

Explainability does not carry a causal claim and does not assure correctness; an incorrect prediction can also produce a convincing explanation, and the explanation may then legitimise the error. Rudin argues that for high stakes decisions, interpretable models should be used from the outset instead of attempting to explain black box models [8]. In the clinical context this debate is not closed.

5.3. Reporting standards

The auditability of a claim depends on compliance with the relevant reporting standard.

Standard	Scope of use	Reference
TRIPOD+AI	Development and validation of clinical prediction models	BMJ 2024;385:e078378
PROBAST+AI	Risk of bias assessment in prediction model studies	BMJ 2025;388:e082505
DECIDE-AI	Early stage clinical evaluation of decision support systems	Nat Med 2022;28:924-933
CONSORT-AI	Randomised trials testing an artificial intelligence intervention	BMJ 2020;370:m3164
SPIRIT-AI	Protocol stage of the same trials	BMJ 2020;370:m3210

The practical use of these lists is to ask how many items on the relevant checklist have been answered when a system is proposed. Every unanswered item is recorded as an item of uncertainty.

5.4. Designing human oversight

Human oversight is a set of concrete design decisions rather than a slogan. Threshold selection is a clinical decision, and the acceptable balance between sensitivity and specificity is determined by the clinical team rather than the model. The model should be able to decline to predict where its confidence interval is wide. Where a patient unlike the training population is encountered, the system should not predict silently. The circumstances under which a decision is escalated should be documented in advance.

6. Regulatory framework

6.1. Is this a medical device?

Software placed on the market by a manufacturer for a diagnostic or therapeutic purpose falls within the scope of medical device legislation. In the European Union the assessment is made under the Medical Device Regulation (EU) 2017/745, and in particular under Rule 11, which classifies software [9]. A negative answer does not remove liability; it changes the layer of legislation from which liability arises.

6.2. European Union

The European Union Artificial Intelligence Act, Regulation (EU) 2024/1689, entered into force on 1 August 2024. The Digital Omnibus on Artificial Intelligence, Regulation (EU) 2026/1744, was published on 24 July 2026 and entered into force on 27 July 2026 [13, 14].

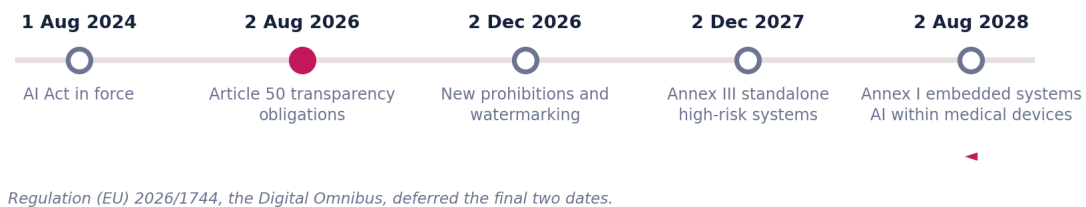


Figure 6. The EU Artificial Intelligence Act timetable as amended in July 2026.

Through that instrument, obligations for standalone high risk systems under Annex III were deferred to 2 December 2027, and obligations for Annex I systems embedded in products already covered by product safety legislation were deferred to 2 August 2028. Artificial intelligence components within medical devices fall into the Annex I group. Transparency obligations under Article 50 were not deferred and applied from 2 August 2026 [14].

6.3. United States

The United States Food and Drug Administration has granted marketing authorisation to approximately 1,450 artificial intelligence enabled devices since 1995. Most are in radiology, cardiology and neurology, and the majority followed the 510(k) pathway. To date no device powered by generative artificial intelligence or by a large language model has been authorised [3]. The agency states explicitly that the list is not a comprehensive inventory; the totals vary with the date of download and should not be presented as an exact count.

The lawful updating of learning systems is governed by the predetermined change control plan mechanism set out in the final guidance issued in December 2024 [15].

6.4. Jurisdictions with no dedicated artificial intelligence statute

Many jurisdictions have no statute regulating artificial intelligence directly. This does not mean that liability is undefined. Liability is established through the combined application of existing layers of legislation, and three layers recur across jurisdictions.

Layer	Relevance to artificial intelligence in health
Medical device legislation	Whether the software qualifies as a device, registration and market placement conditions
Data protection law	Health data is commonly a special category of personal data, with conditions on processing and on automated decisions
Medical malpractice regime	Liability of the clinician and the institution, and the scope of informed consent

Legislation in this area is subject to change. The assessment in this section is current as of September 2026 and does not constitute legal advice.

7. Bridge to the workshop

With the spread of generative artificial intelligence tools, the cost of writing code has fallen markedly. A researcher who has a clinical idea but no knowledge of Python can produce a working decision support system within a few hours. While the cost of writing has fallen, the cost of verification has not.

The workshop tests that possibility directly. Participants write no code across the six notebooks. Each step supplies a prompt written in plain clinical language; the participant passes it to a generative artificial intelligence tool, pastes the code it returns into the notebook and runs it. No knowledge of Python is required.

Each prompt ends with the expected result: which name must hold which piece of information once the code has run. The check cell that follows the paste tests exactly that and reports anything missing. The participant returns the report to the tool and has the code regenerated. Responsibility for verification therefore stays with the user, and that is the habit the workshop sets out to teach.

The code written carries forward from notebook to notebook. At the end of each notebook every step produced so far is collected into a single block and pasted into the first cell of the next notebook. By the end of the sixth notebook the accumulated code is a complete clinical decision support system running behind a web interface.

Generative tools exhibit three typical failures in the health context. The first is the fabricated clinical threshold, where a cut-off value with no traceable source is written into the code in confident language. The second is silent leakage, where the generated code inadvertently mixes test data into the learning process. The third is the hallucinated citation, where a reference to a non-existent article is the most dangerous output precisely because it is formally correct. The check cells and the expected result sections in the workshop target these three risks.

A short Python note follows each step. The notes assume nothing and explain the structure the participant has just seen: what a library is, what a table is, what a function and a docstring do, the difference between printing and returning, how dictionaries and loops are read. The aim is not to have participants write code but to make the code that arrives readable to them.

Participants choose one of six prepared datasets. Three hold real clinical data in tabular and signal form: the MIMIC-IV demo intensive care records, the Breast Cancer Wisconsin set and the MIMIC-IV-ECG demo recordings. Two are openly accessible medical image collections. The sixth is clinical text, and because no note collection is available without credentialing, data is generated on that route. None requires a password and all are reached directly through a browser.

Whichever is chosen, the loading step produces a result of the same shape: a table holding a patient identifier and a target. That shared shape lets every later step proceed identically regardless of data type. The electrocardiography set covers the same patients as the intensive

care set, so a participant taking that route predicts the same outcome from the signal instead, and the two approaches become comparable.

8. Closing

Five conclusions are intended to be carried away. A decision support system does not diagnose; it allocates clinical attention. Building one passes through thirteen stages, and most errors are introduced in the first five, namely in defining the decision moment and preparing the data. A performance claim without external validation remains only a claim. Explainability is a powerful instrument for finding errors and is not an assurance of correctness. Generative artificial intelligence writes code but assumes no responsibility.

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Acknowledgement

This session was delivered within the Technology and Artificial Intelligence Literacy Training in Health Sciences project supported by TÜBİTAK and hosted by Akdeniz University. The author thanks the project coordinators and the organising committee for the invitation and for the organisation of the programme.

The complete lecture and workshop material is published as an open repository at github.com/utkukose/cdss-genai-NB-lecture